

School-Based Mentoring for Adolescents: A Systematic Review and Meta-Analysis

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Abstract

Objectives: To evaluate the impact of school-based mentoring for adolescents (11–18 years) on academic performance, attendance, attitudes, behavior, and self-esteem. **Method:** A systematic review and meta-analysis. The authors searched 12 databases from 1980 to 2011. Eight studies with 6,072 participants were included, 6 were included in meta-analysis. Studies were assessed using the Cochrane Collaboration Risk of Bias Tool. **Results:** Across outcomes, effect sizes were very small (random effects), and most were not significant. The magnitude of the largest effect (for self-esteem) was close to zero, $g = 0.09$, [0.03, 0.14]. **Conclusions:** The mentoring programs included in this review did not reliably improve any of the included outcomes. Well-designed programs implemented over a longer time might achieve positive results.

Keywords

mentoring, adolescents, systematic review

Introduction

In developed countries, 7.3% of males and 8.3% of females aged 15–19 are neither employed nor in education (Organization for Economic Co-Operation and Development [OECD], 2008). Such young people are likely to be from low socioeconomic backgrounds, to lack social capital, and to have few educational qualifications (Bynner & Parsons, 2002). Young people with behavioral and social problems are at particular risk of poor educational outcomes and expulsion from school.

A United Nations report identified the United Kingdom and the United States of America as two of the worst performing nations on a variety of measures related to child well-being (The United Nations Children's Fund [UNICEF], 2007). In the United Kingdom, the cost of being not in employment, education, or training (NEET) during adolescence may reach £97,000 over a person's lifetime (Sodha & Guglielmi, 2009). In the long-term, NEET adolescents experience greater rates of unemployment, mental health problems, and criminal activity (Church, 2003; Utting, Monteiro, & Ghate, 2007). Boys and ethnic minorities are at greatest risk of negative outcomes (Department for Children Schools and Families [DCSF], 2009).

There are many interventions for at-risk youth that aim to increase school attendance and attainment, including career academies in the United States (and more recently in the United Kingdom), the U.K. New Deal for Young People, and the educational maintenance allowance. Targeted interventions may focus on social skills or bullying in schools (Vreeman & Carroll, 2007; Wilson & Lipsey, 2007). The most specific

treatments for antisocial behavior, such as functional family therapy (Utting et al., 2007) and multisystemic therapy (Littell, Popa, & Forsythe, 2005), may have positive effects, but they often require trained professionals, so they are expensive to implement. Given the number of adolescents at risk of negative long-term outcomes (Sodha & Guglielmi, 2009), social workers and other professionals have sought inexpensive interventions that could be delivered to large parts of the population. Different forms of mentoring have been used to help young people develop positive relationships with adults or older peers. Proponents hope that role models will provide positive reinforcement and help adolescents solve problems, thus promoting positive behavioral and educational outcomes.

Long used by professionals, mentoring is increasingly offered to young people who are hard to reach (Roberts, Liabo, Lucas, DuBois, & Sheldon, 2004; Sipe & Roder, 1999). Community- and school-based mentoring schemes are expanding rapidly, and they are used widely in the United States and United Kingdom. For example, the Big Brothers Big Sisters of America school-based mentoring program quadrupled in size between 1999 and 2006 (Herrera, Grossman,

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Kauh, Feldman, & McMaken, 2007). Compared to professionally delivered interventions, mentoring is inexpensive, and it can be intensive if delivered frequently (Grossman & Tierney, 1998; Hall, 2003; Herrera, Sipe, McClanahan, & Arbretton, 2000; Roberts et al., 2004). Mentoring shares common features with positive youth development programs that aim to build on young people's strengths rather than focus on their problems (Blum, 2003). Like other positive youth development programs, mentoring aims to build "competence, confidence, character, connection, and caring" (Lerner, Almerigi, Theokas, & Lerner, 2005). Many positive youth development programs include a mentoring component as one of the many activities; this review aims to measure the effects of mentoring itself, rather than the combined effect of a wider program.

Though increasingly popular, previous reviews show that the effects of mentoring are not well understood (Herrera et al., 2000). Mentoring programs vary in the amount of training and supervision given to mentors. Most programs aim to support mentors by contacting them at least once a month but about 10% provide no training (Sipe & Roder, 1999). First meetings usually occur in an organization's center, and they may range in frequency. Pairs may decide what to do; activities can include sports, social events, meals, or conversation (Jekielek, Moore, & Elizabeth, 2002).

Mentoring aims to strengthen attachment, improve social competence, and increase social capital by introducing new connections (Boeck, Fleming, & Kemshall, 2006; Philip & Spratt, 2007). A strong attachment with a caring adult may help build resilience (Jenkins, 2008). Several studies have found that social support predicts healthy behavior and academic achievement (Blum, 2005; Blum & Ireland, 2004; Dubow, Tisak, Causey, Hryshko, & Reid, 1991; Hall, 2003), and improved academic performance may be a distal outcome of mentoring (Rhodes, Grossman, & Resch, 2000). Some studies suggest that mentoring has modest effects on attendance (Hall, 2003), academic performance, employment, and behavior (DuBois, Holloway, Valentine, & Cooper, 2002).

School-based mentoring is often administered by established mentoring charities like Big Brothers Big Sisters that recruit and screen community volunteers, matching them with young people. Other programs may be organized by schools or by social workers. School-based mentoring forms a large part of new mentoring schemes (Jekielek et al., 2002) and may have advantages over other community schemes. By comparison, organization and procedures may be simplified; some meetings can take place on the school site, which is thought to be safe. Teachers may be well placed to select young people who are likely to benefit from mentoring, and teachers can check that meetings occur and meet participants' needs (i.e., that programs are well implemented). Mentors may be older volunteers including older adolescents serving as mentors to people in the same school. Adolescence is characterized by increased importance of peer relationships, and several studies identify negative effects of "deviant" peer relationships that undermine prosocial behavior at school and in communities (Dodge, Dishion, & Lansford, 2006a, 2006b). Strong mentor relationships

with adults or older students may replace or counterbalance negative influences.

There have been several reviews of youth mentoring programs overall (DuBois et al., 2002; Tolan, Henry, Schoeny, & Bass, 2008), which found small positive effects for behavior and academic outcomes, but no reviews have examined school-based mentoring in particular. There have been several large studies of school-based mentoring reported since the last reviews, thus a systematic review is needed to synthesize available studies.

Method

This review and meta-analysis describes the effects of school-based mentoring for adolescents on academic performance, attendance, attitudes, behavior, and self-esteem.

Systematic reviews are the highest level on the hierarchy of evidence for evaluating the effectiveness of interventions (Sackett, Richardson, Rosenberg, & Haynes, 1997). They ask clearly defined questions, their methods are transparent and replicable, and they use rigorous procedures to minimize reviewer bias and to draw attention to potential sources of bias in included studies. Systematic searches aim to gather *all* studies that relate to the population and intervention of interest. Structured literature searches and analyses facilitate updates using the same procedures. Thus, the eight studies in this review are the best available evidence about the effectiveness of school-based mentoring.

This report conforms to PRISMA guidelines (Moher, Liberati, Tetzlaff, & Altman, 2009) for the reporting of systematic reviews and meta-analyses. The protocol and data extraction forms are available from the authors.

Eligibility

Studies were eligible if they evaluated school-based mentoring for people aged 11–18 years compared to a contemporaneous control (including no treatment or an alternative treatment). Both randomized and quasi-experimental designs were included. Mentors were older volunteers (i.e., not professionals). Searches were limited to studies reported since 1980 and were not restricted by country.

Data for the following outcomes were extracted (Figures 3–7):

- Academic achievement (e.g., grades).
- School attendance.
- Attitude (e.g., toward school or family).
- Behavior (e.g., substance use).
- Self-esteem.

Search Methods for Identification of Studies

A systematic literature search was conducted through March 2011 to identify all published and unpublished studies meeting the above criteria (Figure 1). The following databases were searched (using the Ovid platform for Medline and PsycINFO): Applied Social Sciences Index and Abstracts; Australian

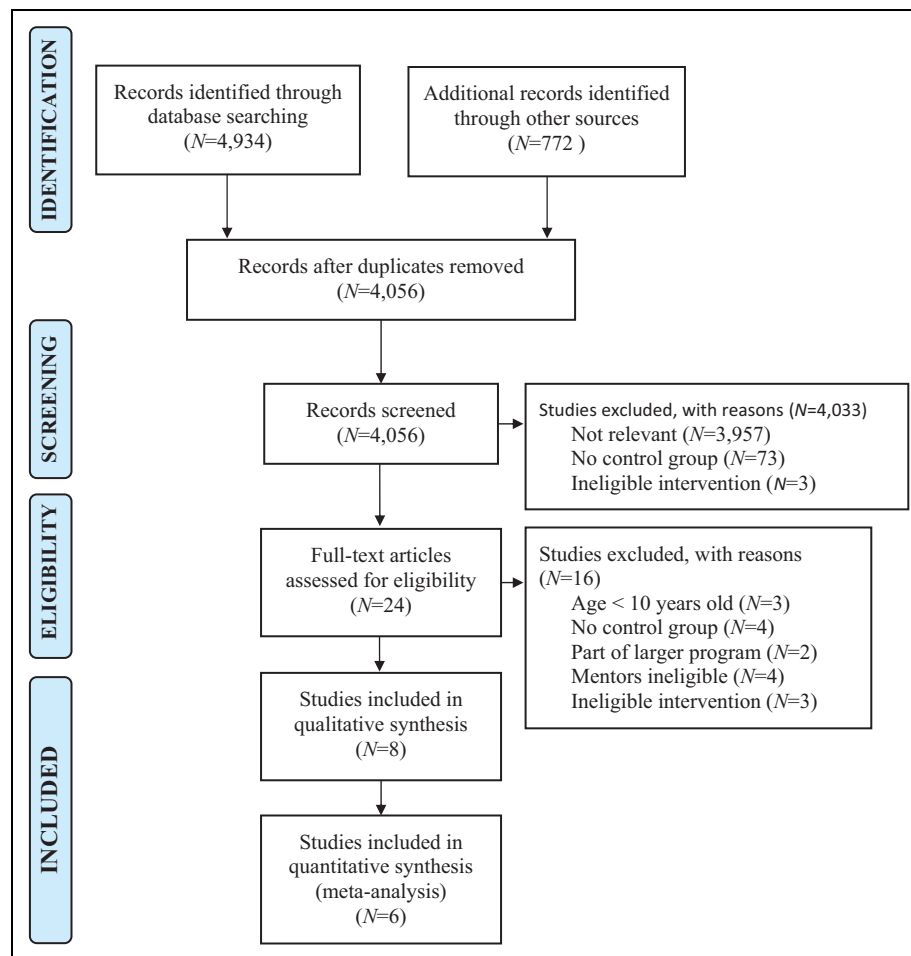


Figure 1. PRISMA flowchart.

Education Index; British Education Index; Cochrane Central Register of Controlled Trials in the Cochrane Library; Database of Abstracts of Reviews of Effectiveness, Educational Resources Information Center; MEDLINE; PsycINFO; Social Sciences Citation Index. For gray literature, the following were searched: Dissertation Abstracts, GreyNet, and Open Sigle. Reference lists of articles identified through database searches were examined to identify further relevant studies. As school-based mentoring is a relatively recent development, searches were limited to studies reported from 1980.

The following terms were used for MEDLINE (these were modified for other databases; terms are available from the authors on request):

1. exp Mentors/
2. exp Helping Behavior/
3. exp Interpersonal Relations/
4. mentor*.mp
5. coach*.mp
6. (interpersonal or helping) adj relationship*.mp
7. role adj model*.mp
8. 1 or 2 or 3 or 4 or 5 or 6 or 7
9. exp Child Behavior Disorders/
10. exp Juvenile Delinquency/
11. exp Risk-Taking/
12. exp Conduct Disorder/
13. exp Attention Deficit and Disruptive Behavior Disorders/
14. (emotion* and behav* and difficult*).mp
15. ("anti social" or antisocial) and (behavior* or behaviour*).mp
16. (behav* or conduct) and (problem* or disorder*).mp
17. Risk-taking.mp
18. truan*.mp
19. dropout*.mp
20. (drop* adj out).mp
21. delinquen*.mp
22. devian*.mp
23. EBD.mp
24. (exclu* or expel* or expulsion or suspen*).mp
25. 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24
26. exp Schools/
27. school*.mp
28. (academy or academies).mp
29. 26 or 27 or 28
30. 8 and 25 and 29

	Adequate sequence generation?	Allocation concealment?	Blinding?	Incomplete outcome data addressed?	Free of selective reporting?
Aseltine 2000	?	?	+	+	?
Bernstein 2009	+	+	+	+	?
Herrera 2007	+	?	+	+	?
Karcher 2008	?	?	+	+	?
LoSciuto 1996	?	?	+	+	+
McPartland 1991	+	+	+	+	?
Portwood 2005	+	+	+	+	+
Whiting 2007	+	+	+	+	+

Figure 2. Risk of bias.

Study Selection

One author (S.W.) screened abstracts for eligibility. Full texts were examined by both authors, and the included and excluded studies were agreed through discussion. Data were extracted by one author (S.W.) and checked by a second (E.M-W.); discrepancies were resolved through discussion.

Assessment of Bias

Studies were assessed using the Cochrane Collaboration Risk of Bias Tool (Higgins, Altman, & Green, 2009). Each study was rated by two authors as *high*, *low*, or *unclear* for risk of bias due to sequence generation, allocation concealment, blinding (of participants, providers, and assessors), selective outcome reporting, and incomplete data. Discrepancies were resolved through discussion.

Data Management and Analysis

Information about participant and intervention characteristics were extracted from published and unpublished reports. When outcome data were reported in more than one format, raw values (e.g., mean and standard deviation) were extracted rather than calculated effect sizes (e.g., odds ratio). When data were not available in published reports, authors were contacted for missing information.

For all outcomes, we calculated Hedges g and 95% confidence intervals (CI), and overall effects were weighted by the inverse of variance using comprehensive meta-analysis. As trials may test different program effects across studies, we used random effects to calculate the average treatment effect (Riley, Higgins, & Deeks, 2011). (Positive values indicate superiority of the intervention, and groups are equivalent when $g = 0$.)

Statistical heterogeneity was assessed by visual inspection of forest plots, by performing the χ^2 test (and assessing the p value) and by calculating the I^2 statistic, which describes the percentage of observed heterogeneity that cannot be explained by chance. If the p value was less than .10 and I^2 exceeded 50%, we considered heterogeneity to be substantial (Higgins et al., 2009). To assess the possibility of small study bias, random-effects estimates were compared to fixed-effect estimates (Higgins et al., 2009).

When a single study provided multiple measures of the same outcome, a combined effect was calculated for the outcome. For outcomes reported as F tests, we assumed a pre-post correlation of .5. Data that could not be used to calculate an effect size are reported separately.

Results

Trial Flow

The literature search identified 4,056 unique citations, of which 24 were retrieved for further analysis. Eight studies were included (Figure 1) and studies that did not meet the inclusion criteria are described in the table of excluded studies (Table 1).

Participant Characteristics

Studies assigned 6,072 participants (range 82–2,573) and their characteristics are described in the table of included studies (Table 2). Five compared mentoring to treatment as usual (Bernstein, Rappaport, Olsho, Hunt, & Levin, 2009; Herrera et al., 2007; Karcher, 2008; Portwood et al., 2005; Whiting & Mallory, 2007); others tested mentoring as an adjunct to other interventions (Aseltine, Dupre, & Lamlein, 2000; LoSciuto et al., 1996; McPartland & Nettles, 1991).

Participants were mainly in middle school, with an average age of 11 years. Just over half of the participants in the review are female. Most participants were from low-income families; for example, in the two largest studies, 86% of participants (Bernstein et al., 2009) and 69% of participants (Herrera et al., 2007) were eligible for free or reduced school meals. Across trials, about 34% of participants were Black, 31% Hispanic, and 24% White. All studies were conducted in the United States.

Intervention Characteristics

Mentoring was provided by nonprofit or community-based organizations. Where characteristics about mentors were reported, about 70% were female. In the largest studies, most

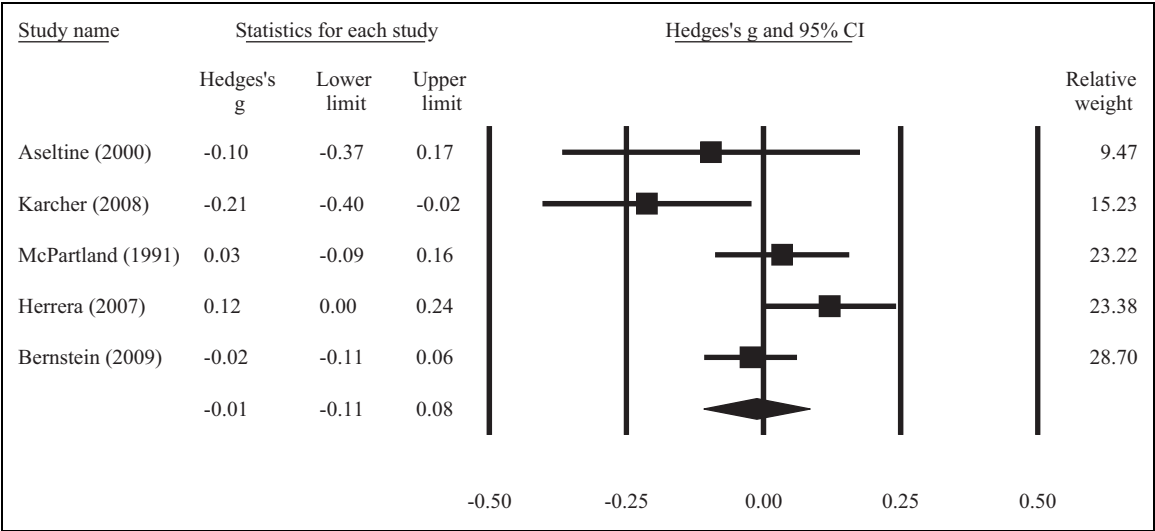


Figure 3. Academic achievement.

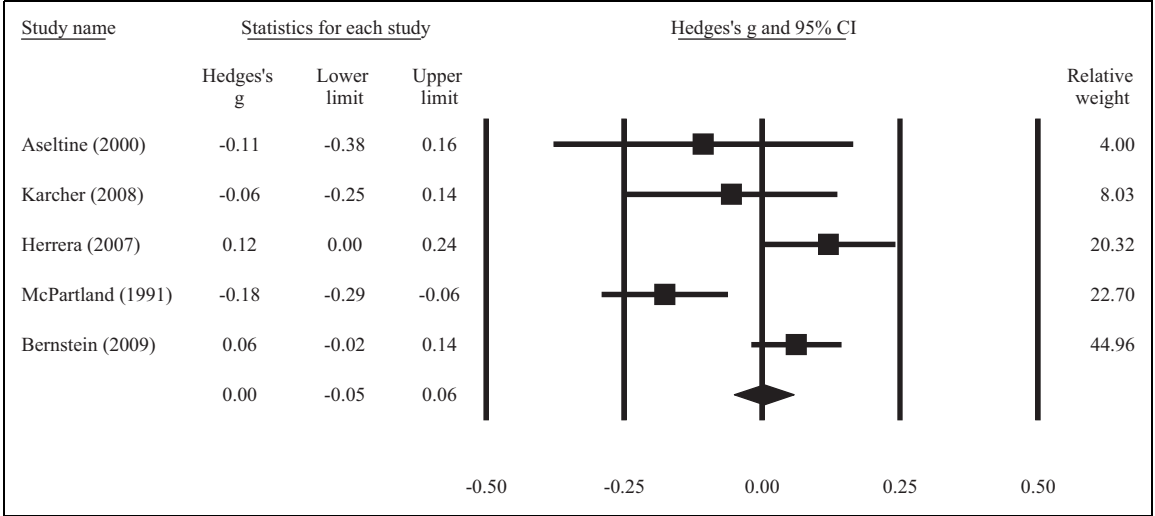


Figure 4. Attendance.

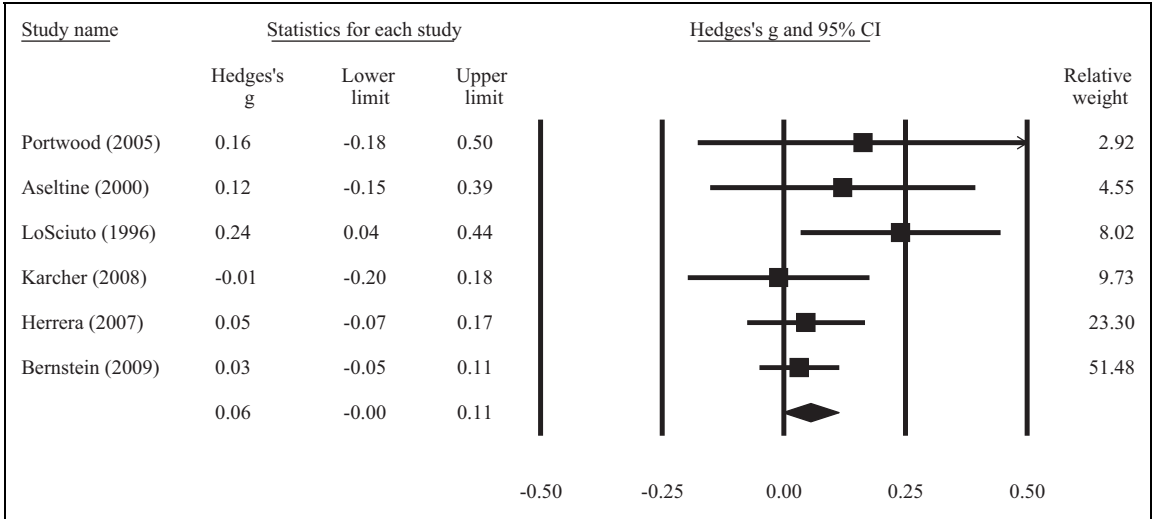


Figure 5. Attitude.

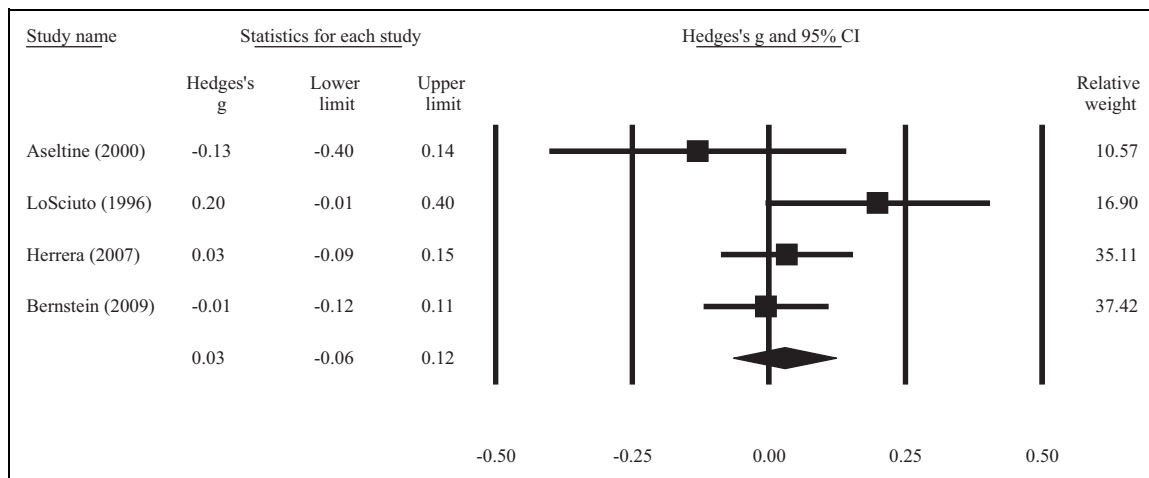


Figure 6. Behavior.

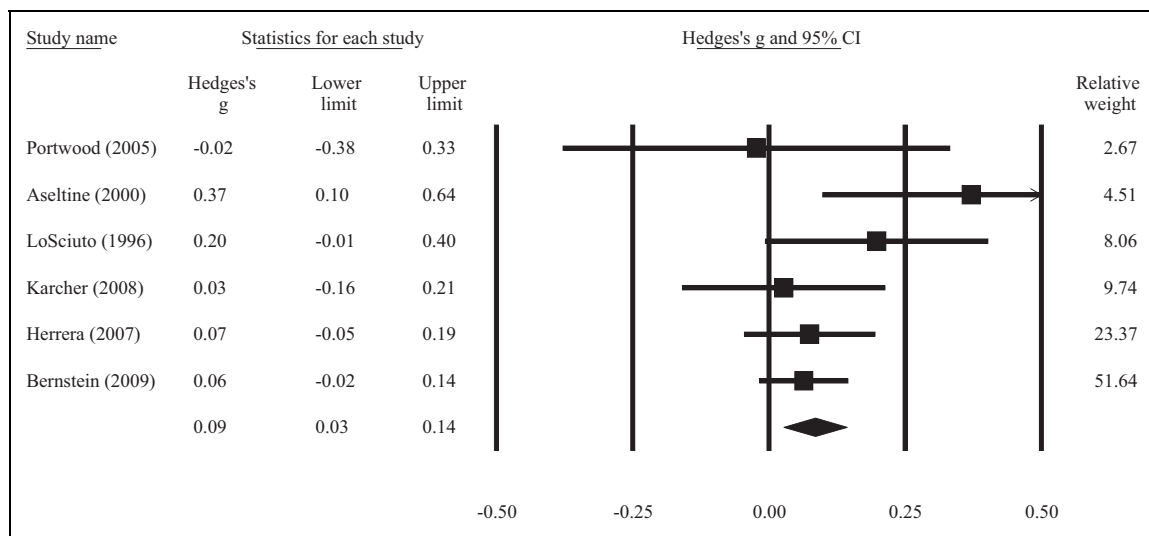


Figure 7. Self-esteem

were White, though the majority of mentors in other studies were Latino (Karcher, 2008) and Black (LoSciuto et al., 1996). In six studies, at least half the mentors were under 25; two programs used older adults (Aseltine et al., 2000; LoSciuto et al., 1996). Training varied in length from 1 hr to 2 days (LoSciuto et al., 1996).

Mentoring was usually on-site but not always; some included outings (Portwood et al., 2005) or community visits (Aseltine et al., 2000; LoSciuto et al., 1996). Meetings included casual conversation, and sometimes academic work. Dosage varied in length from the usual amount of 1 hr/week (Bernstein et al., 2009; Karcher, 2008; Portwood et al., 2005) to 6 hr/week (Aseltine et al., 2000; LoSciuto et al., 1996). One program lasted only 2 months (Aseltine et al., 2000), another lasted nearly 6 months (Bernstein et al., 2009). In one study, some relationships lasted more than a year (Herrera et al., 2007); other studies occurred over several years, but the average

dosage was unclear (McPartland & Nettles, 1991; Whiting & Mallory, 2007).

Potential Sources of Bias

Studies were assessed using the Cochrane Risk of Bias tool (Higgins et al., 2009) and a summary was generated (Figure 2).

To evaluate the effectiveness of an intervention, randomization to intervention and control groups is the best way to minimize selection bias. Five studies including 5,232 participants (86%) were randomized controlled trials (RCTs; Aseltine et al., 2000; Bernstein et al., 2009; Herrera et al., 2007; Karcher, 2008; LoSciuto et al., 1996), three studies were at high risk of selection bias as they were nonrandomized. One used alternating assignment (Whiting & Mallory, 2007) and another matched participants on gender, race, grade, and socioeconomic status (Portwood et al., 2005). In the third,

Table 1. Excluded Studies

Study	Reason for Exclusion
Adler, Cragin, and Searls (1995)	Mentoring undifferentiated within other program
Cave and Quint (1990)	Ineligible intervention
Cavell and Hughes (2000)	Age <10 years old
Converse and Lignunaris/Kraft (2008)	Mentors ineligible
Gibson (2003)	Age <10 years old
Higgins, Furano, Toso, and Branch (1991)	Mentoring undifferentiated within other program
Jackson (2002)	No control group
Johnson, Holt, Bry, and Powell (2008)	Mentors ineligible
Maxfield, Schirm, and Rodriguez-Planas (2003)	No control group
Rose and Jones (2007)	No control group
Shiner, Young, Newburn, and Groben (2004)	Ineligible intervention
Slicker and Palmer (1993)	Mentors ineligible
Vidourek, King, Davis, and McClellan (2002)	Age <10 years old
Waits (2010)	No control group
Wright, Lindsay, Livingstone, Shepherd, and Duku (2007)	Mentors ineligible
Yampolskaya, Massey, and Greenbaum (2006)	Ineligible intervention

comparison group members were chosen from students not eligible for the program, and the analyses controlled for baseline differences (McPartland & Nettles, 1991).

Selection bias can be introduced in RCTs if the allocation sequence is not concealed, thus removing unpredictability (Pildal et al., 2007). Only one study adequately concealed the allocation sequence (Bernstein et al., 2009). Allocation concealment was not reported for any of the other RCTs and was inadequate per se in all nonrandomized studies.

Blinding reduces the likelihood of response bias and reduces the possibility of biased coding by assessors (Devereaux, Choi, El-Dika, Bhandari, & Montori, 2004). Given the nature of the intervention, it was not possible to blind participants or mentors. Studies could have blinded some outcome assessors, but none reported doing so.

Attrition was less than 10% overall, though studies were not consistent in reporting these numbers. Three studies (Bernstein et al., 2009; Herrera et al., 2007; Karcher, 2008) analyzed participants in the groups to which they were randomized irrespective of whether they dropped out or were lost to follow-up (i.e., intention-to-treat). Five studies only analyzed those participants who gave both pre- and posttest data and are at high risk of bias due to incomplete outcome data.

When nonsignificant outcomes are not reported or analyzed, results of meta-analyses can be distorted. To identify selective outcome reporting bias, this review was *not* limited to studies that reported particular outcomes (Hutton & Williamson, 2000). Three studies did not report all outcomes measured, which indicates risk of bias (LoSciuto et al., 1996; Portwood

et al., 2005; Whiting & Mallory, 2007); neither protocols nor trial registrations were identified for the others.

Effects of Interventions

Six studies reported posttreatment data that could be combined for meta-analysis; these studies included 88% of participants in the review. Follow-up data were not combined because periods to follow-up were very different, but effects are reported for each study.

Five forest plots were constructed, one for each of the eligible outcomes. There was one significant effect, but the magnitude of all effects were clinically unimportant (none of the combined results were larger than $g = 0.1$). Results were strongly influenced by two large studies (Bernstein et al., 2009; Herrera et al., 2007), which contributed 71% of participants in the review.

Academic Achievement

Measures of academic achievement included promotion, grades, completion of assignments, standardized aptitude test scores, and classroom assessment techniques scores. These were reported in seven studies (88%), contributing 4,769 (89%) participants, though data could be combined for only four studies (50%).

At posttreatment, there was no difference between groups in academic achievement, $g = -0.01 [-0.11, 0.08]$, though there was some statistical heterogeneity across studies ($I^2 = 59\%$; $\chi^2 = 9.81, p = .04$). Two studies could not be pooled; they reported a decrease in grade levels (Whiting & Mallory, 2007) and no effect (Portwood et al., 2005). There were no consistent differences between the randomized and nonrandomized studies. Taken together, data that could not be analyzed confirm the result of the meta-analysis, that is, these mentoring programs did not affect academic achievement in the short term.

One study (Aseltine et al., 2000) reported no difference at 6-month follow-up, $g = 0.12 [-0.17, 0.40]$ and another (Herrera et al., 2007) reported no differences after 15 months, $g = 0.01 [-0.09, 0.14]$. A 10-year follow-up (Whiting & Mallory, 2008) found no effect on high school graduation rates or entry into postsecondary education.

Attendance

Measures of attendance included self-reported truancy and school records.

At posttreatment, there was no overall difference between groups, $g = -0.02 [-0.14, 0.10]$ but statistical heterogeneity was significant ($I^2 = 75\%$; $\chi^2 = 16.3, p = .003$). Of the studies that could not be pooled, both reported small-to-modest improvements in attendance (LoSciuto et al., 1996; Whiting & Mallory, 2007).

At 6-month follow-up, one study (Aseltine et al., 2000) reported no difference between groups, $g = -2.31 [-2.68,$

Table 2. Included Studies

Study	Participants					Mentors			
	Dose	Number	Grade	% F	Race	Number	Age	% F	Race
Aseltine, Dupre, and Lamlein (2000)	1 p/w, 2 m	505	6	NA	NA	NA	65+	NA	NA
Bernstein, Rappaport, Olsho, Hunt, and Levin (2009)	1 p/w, 6 m	2,573	4–8	53	Black 41% Hispanic 31%	974	82.5% >18	72	Black 29% Hispanic 10%
Herrera, Grossman, Kauh, Feldman, and McMaken (2007)	1 p/w, 5 m	1,139	4–9	54	Black 18% Hispanic 23%	554	52% > 18	72	Black 8% Hispanic 6%
Karcher (2008)	1 p/w, 3 m	525	5–8	66	Black 8% Hispanic 79%	292	70% College students	73	Black 5% Hispanic 54%
LoSciuto, Rajala, Townsend, and Taylor (1996)	NA	729	6	53	Black 52% Hispanic 9%	NA	65+	NA	NA
McPartland and Nettles (1991)	2 p/w, 24 m	311	6	NA	NA	NA	NA	NA	NA
Portwood et al. (2005)	1 p/w, 8 m	208	4–12	52	Black 16% Hispanic 8%	211	46	80	NA
Whiting and Mallory (2007)	1 p/w, 24 m	82	5–6	49	Black 97%	30	College students	NA	NA

–1.95]. At 15 months, another (Herrera et al., 2007) reported that youth from the mentored group were less likely to skip school, but combined effects for reported measures of attendance were not significant, $g = 0.09$ [–0.03, 0.22].

Attitude

Attitudinal measures were the most diverse, including relationships with other adults, peers and teachers, school and community connectedness, personal initiative, and future orientation. Measures were both school- and self-reported. Attitudes were reported in seven (88%) studies, contributing 5,021 (94%) participants, but only six could be pooled. Due to lack of information about dropout, the trial reporting the most positive effect (LoSciuto et al., 1996) may be slightly overweighted in the analysis, but this did not affect any conclusions.

At posttreatment, there was no difference between groups, $g = 0.06$ [–0.00, 0.11] and heterogeneity was neither important nor significant ($I^2 = 37\%$; $\chi^2 = 7.94$, $p = .16$).

One study that could not be included in the meta-analysis reported that some outcomes were significantly better in the intervention group (Whiting & Mallory, 2007). Together, the results suggest that these mentoring programs did not consistently affect attitudes in the short-term, and overall effects would remain trivial if the missing studies had been included.

After 6 months, one trial found no effect (Aseltine et al., 2000). Another trial reported no effect after 15 months, $g = 0.05$ [–0.08, 0.17] (Herrera et al., 2007).

Behavior

Behavioral outcomes included self- and school-reported misconduct, delinquent behavior, and drug use.

At posttreatment, there was no difference between groups, $g = 0.03$ [–0.06, 0.12], and heterogeneity was neither

important nor significant ($I^2 = 31\%$; $\chi^2 = 4.34$, $p = .23$). One study could not be pooled for lack of information about group size, variance, and outcome definition (Whiting & Mallory, 2007), but it reported an “improvement in discipline.”

At 6 months, one study (Aseltine et al., 2000) reported significant improvements in behavior, $g = 1.79$ [1.45, 2.13], but posttreatment differences in another study (Herrera et al., 2007) were not observed at 15 months, $g = 0.01$ [–0.12, 0.14].

Self-Esteem

Measures of self-esteem were all self-reported, and we included psychological well-being and scholastic efficacy in this analysis.

At posttreatment, there was a trivial but statistically significant difference between groups, $g = 0.09$ [0.03, 0.14], and heterogeneity was neither important nor significant ($I^2 = 22\%$; $\chi^2 = 6.42$, $p = .27$). Data from one trial could not be included (Whiting & Mallory, 2007); the authors reported the effect was significant but did not provide summary statistics (e.g., means or standard deviations).

After 6 months, one trial reported that effects were no longer significant (Aseltine et al., 2000) and another trial reported no effect, $g = 0.04$ [–0.08, 0.17], after 15 months (Herrera et al., 2007). A 10-year follow-up (Whiting & Mallory, 2008) found that improvements were not maintained and there were no significant differences between the groups.

Subgroup Analyses

We planned to conduct subgroup analyses to determine whether outcomes were related to characteristics of the interventions or participants, including age and gender of mentees, and duration and quality of relationship. Data could not be

Table 3. Outcomes After One School Year

Outcome	Number of Trials (% of Total)	Number of Participants (% of Total)	g [95% CI] Random Effects (Fixed Effect)	Heterogeneity I^2 ; χ^2 (p Value)
Academic Achievement	5 (63%)	4,420 (92%)	−0.01 [−0.11, 0.08] (0.00 [−0.06, 0.06])	59%; $\chi^2 = 9.81$, $p = .04$
Attendance	5 (63%)	4,815 (93%)	−0.02 [−0.14, 0.10] (0.00 [−0.05, 0.06])	75%; $\chi^2 = 16.3$, $p < .01$
Attitudes	6 (75%)	4,949 (99%)	0.06 [−0.00, 0.11] (0.01 [−0.03, 0.06])	37%; $\chi^2 = 7.94$, $p = .16$
Behavior	4 (50%)	4,347 (98%)	0.03 [−0.06, 0.12] (0.03 [−0.05, 0.10])	31%; $\chi^2 = 4.34$, $p = .23$
Self-Esteem	6 (75%)	4,937 (99%)	0.09 [0.03, 0.14] (0.03 [−0.06, 0.12])	22%; $\chi^2 = 6.42$, $p = .27$

Note. CI = confidence interval

pooled for analysis across studies, and individual effects are reported below.

Two studies examined the effect of age. One study that included participants aged 9–15 years found improvements in attendance for older students but not the younger ones (Herrera et al., 2007); another found a contradictory result with lower truancy among those under 12 (Bernstein et al., 2009).

Gender effects were analyzed in four studies (Bernstein et al., 2009; Bogat & Liang, 2005; Herrera et al., 2007; Karcher, 2008; Portwood et al., 2005). Some indications of iatrogenic effects for boys were found in attitudinal outcomes for two studies (Bernstein et al., 2009; Karcher, 2008). Effects on girls were contradictory, with one study (Herrera et al., 2007) reporting an improvement in academic outcomes and another (Bernstein et al., 2009) reporting a small decrease. A similar contradiction was found in self-esteem outcomes; two studies found girls improved (Bernstein et al., 2009; Karcher, 2008), while one reported that boys improved more than girls (Portwood et al., 2005). Given that there is little evidence of any overall effects and there is evidence of selective outcomes reporting, these reported differences may be best explained by chance.

Quality and duration of relationships were reported in two studies (Herrera et al., 2007; LoSciuto et al., 1996). One found that improved mentor training, support, and resources were strongly associated with both relationship quality and relationship duration (Herrera et al., 2007). The other found that those who had “high-quality” mentoring did better on all outcome measures (LoSciuto et al., 1996). This could suggest a causal mechanism, but the relationship could be explained by a common cause.

Discussion and Applications to Social Work

This review systematically identified and analyzed results from all known studies of school-based mentoring. Though individual study reports identify some positive outcomes, the totality of the evidence suggests that these interventions did not achieve their intended results. Evidence from 6,072 young people in eight studies suggests that mentoring did not improve

academic achievement, attendance, behavior, and psychological outcomes. Some studies reported differences by gender and age, but there were no consistent results to suggest that any subgroups were affected or affected differently.

The results of this review are statistically robust. Using a fixed-effect model, none of the combined results are meaningfully different from those presented above (see Table 3). The only statistically significant effect (self-esteem) became insignificant, which did not affect the conclusion that these interventions were ineffective.

The programs tested were well established and representative of everyday practice. Many participants received free school meals, and some had a history of behavioral problems. Though not limited to young people with severe problems, these results may generalize to target populations for most mentoring programs.

This review includes the best available evidence, and the two largest studies (60% of participants in the review) were randomized trials. Other studies in the review were at relatively higher risk of bias, which would be more likely to overestimate the effects of mentoring. Across studies, there was no information about blinding, and the studies may have been vulnerable to performance bias or detection bias. Data were missing in most studies, either due to selective attrition or selective outcome reporting. For example, poorer-performing students were dropped from the intervention group in one study (McPartland & Nettles, 1991); this may have occurred in other studies in which participants who continued were keen to do well. To highlight potential limitations in the internal (e.g., due to missing data) and external validity (e.g., due to small samples) of each analysis, the number of studies and participants included is shown in Table 3 (Mayo-Wilson & Montgomery, 2010).

Despite the risk of bias, the overall results show no discernable differences between groups and some uncertainty about effects. We conducted a highly sensitive electronic literature search for studies of mentoring in schools, but a more sensitive gray literature search might identify unknown studies, and further studies could change these conclusions. We did not conduct tests for publication bias because there were few included studies and no overall effects.

To account for variation across studies, we used the random-effects model for meta-analysis, which estimates the mean size of effects that may differ across interventions or populations. In some analyses, statistical heterogeneity was very high, and we were unable to explain why the results differed. However, evidence of positive effects was largely limited to relatively small effect sizes reported for a subset of the outcomes measured. Even considering these differences across studies, there is little evidence to support the effectiveness of the programs tested.

The external validity of these findings is limited insofar as all included studies were conducted in the United States. Studies included young people of various ages. Though the mean age in each study fell within our prespecified inclusion criteria, the results may be influenced by the inclusion of younger participants. These results, which are largely based on studies of participants around 11 years old, may not generalize to older adolescents.

Implementation

Details about the implementation of interventions are routinely underreported in trials of complex interventions (Mayo-Wilson, 2007). It is clear that the programs and participants varied across the included studies, but information about the delivery of mentoring was limited. Consequently, we could not identify program components that might improve or diminish effects. Statistical heterogeneity could have been related to a number of factors.

Mentors were always older than mentees and ranged from under 18 (Herrera et al., 2007) to more than 65 years old (Aseltine et al., 2000; LoSciuto et al., 1996). In one study, they were mostly nurses (Whiting & Mallory, 2007). Many programs aim to support their mentors by contacting them at least once a month (Sipe & Roder, 1999). In this review, mentors received a 2-day training program with regular supervision (LoSciuto et al., 1996) or as little as a 1-hr orientation (Karcher, 2008). In one large study, many mentors received no training before or during the intervention (Herrera et al., 2007); in another, over 90% mentors received training and ongoing support (Bernstein et al., 2009).

Lack of positive effects might be explained in part by poor program differentiation; that is, if the control group received the intervention or if the intervention was not provided as planned, there might not have been a clinically important difference between the groups. For example, only 86% of the intervention group in the largest study received mentoring and 35% of the control group also received mentoring from another source (Bernstein et al., 2009). Previous studies have suggested that about half of mentoring relationships last only a few months (Rhodes, 2002), which was true of studies in this review. Evaluating the effectiveness of mentoring requires greater understanding of the nature of mentoring relationships and how control groups differ from intervention groups in practice rather than design.

Conclusions

This review is largely consistent with earlier research. Previous reviews have suggested that impacts of mentoring were significant but not large (DuBois et al., 2002; Hall, 2003). Peer networks are particularly important in adolescent development (Furman & Buhrmester, 1992), and deviant peers can have powerful effects (Dodge et al., 2006a, 2006b). Even if at-risk young people have positive relationships with mentors, the benefits of relatively weak but positive relationships may be insufficient to overcome negative peer influences (DuBois et al., 2002; Hall, 2003; Tolan et al., 2008). Previous reviews suggest that “mentoring may be valuable for those at-risk or already involved in delinquency and for associated outcomes,” but there is little evidence to support this theory (Tolan et al., 2008). Reviewing 39 mentoring projects for the U.K. Youth Justice Board, an evaluation found that “those on the mentoring program fared a little worse in terms of reoffending than the national cohorts,” however, almost all of the schemes secured further funding (Tarling, Davison, & Clarke, 2004).

This review found no evidence that mentoring causes harm; but like previous reviewers, we remain concerned by the possibility that *failed* relationships could have iatrogenic effects (Bogat & Liang, 2005; Hall, 2003). Mentoring also aims to improve self-esteem, and it could cause harm if it increases risk taking, for example, leading boys to focus on competition other than academic improvement (Karcher, 2008).

School-based mentoring may be inexpensive compared to other interventions, but it is not free. The costs of the Big Brothers Big Sisters program were approximately \$1,000 per student per year (Herrera et al., 2007). Based on a representative sample of 100 providers, the cost of the U.S. Department of Education’s student mentoring program exceeds \$1,500 per student per year (Bernstein et al., 2009). Moreover, programs for the general population may not reach those young people in greatest need of intervention (Rennison et al., 2005). Unless school-based mentoring can demonstrate positive effects, it represents poor value for money no matter its cost.

As school-based mentoring programs are delivered by many organizations and part of routine services, they may be likely to continue despite limited evidence of effectiveness; more research could examine ways to modify and improve these programs to ensure they are safe and to make them effective. It would be particularly useful to understand more about mechanisms of change. Further trials are required to determine whether there is a mediating relationship between outcomes and the amount of mentoring that participants receive, or if young people who are able to form durable relationships experience better outcomes than their peers in the absence of mentoring. Furthermore, trials could examine prospective moderators; for example, interventions for young people with behavioral difficulties may be more effective than those for the general population (Rennison et al., 2005), and evidence of differential effects might have important implications for program design and recruitment. Larger studies with negative results and multiple replications may be required before schools and

community organizations would consider discontinuing these programs.

Participants in school-based mentoring schemes are often disadvantaged young people receiving interventions from several sources. Proponents hope that mentoring will lead to strong, supportive relationships, but the longest programs in this review lasted only a few months for most young people. It is hard to imagine how an unstructured, low-dose intervention would have clinically important benefits beyond other services in the short term. Mentoring programs that create enduring positive relationships might achieve better results, and further literature reviews might examine ways to improve the duration and quality of mentoring relationships.

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